

① Dataset : Hospital Patient Records

Min support = 50%

Min confidence = 70%

Constraint : Every rule must include fever

Calculate support

$$* \text{fever} = P_1, P_2, P_4, P_5 = 4/5 = 80\%$$

$$* \text{cough} = P_1, P_3, P_4 = 3/5 = 60\%$$

$$* \text{Headache} = P_2, P_3, P_4 = 3/5 = 60\%$$

$$* \text{Fatigue} = P_5 = 1/5 = 20\%$$

2. Frequent Pairs: containing fever

$$* \text{Fever} + \text{cough} = P_1 P_4 = 2/5 = 40\% \rightarrow \text{not frequent}$$

$$* \text{Fever} + \text{Headache} = P_2, P_4 = 2/5 = 40\% \text{ not frequent}$$

$$* \text{Fever} + \text{fatigue} = P_5 = 1/5 = 20\% \rightarrow \text{not frequent}$$

Association rules :

$$\text{Fever} \rightarrow \text{cough} : \text{support} = 2/5 = 40\% \text{ which is below } 50\%$$

$$\text{Fever} \rightarrow \text{Headache} : \text{support} = 2/5 = 40\% \text{ which is below } 50\%$$

* No validation rule exists under the given constraints

Justification: Although fever has 80% support its combinations fever can satisfy the minimum support of 50% regardless of confidence req. of 70%.

Dataset : Online Learning Python

min SUPPORT = 40%

Constraint: Every itemset must contain at least two courses

Step 1: Frequent 1-itemsets

<u>Course</u>	<u>Count</u>	<u>SUPPORT</u>
Py	4	80%
SQL	2	40%
DS	2	40%
ML	2	40%

Step 2: Candidate 2-itemsets

Py, SQL $\rightarrow$ $S1, S4 = 2/5 = 40\%$

Py, DS $\rightarrow$ $S2, S4 = 2/5 = 40\%$

Py, ML $\rightarrow$ $S5 = 1/5 = 20\%$

SQL, DS $\rightarrow$ $S4 = 1/5 = 20\%$

DS, ML $\rightarrow$ none = 0%

Frequent itemsets : 1 {Py, SQL} - 40%

2 {Py, DS} - 40%

Pruning Explanation : The min two course constraint immediately eliminates all 1-itemsets from the need to be considered and therefore reduces the search space and computation cost during frequent itemset mining

3 Dataset: Library Borrowing Records

Transaction	Books Borrowed	Step	Frequent 1-itemset	
		<u>Item</u>	<u>Count</u>	<u>Support</u>
T1	Dalchini, Py	Dalchini	4	80%
T2	Py, DBMS	Py	3	60%
T3	Py, DBMS, Dalchini	DBMS	3	60%
T4	Dalchini, DBMS	AI	1	20%
T5	Dalchini, AI			

Step 2: Candidate Itemsets

<u>Itemset</u>	<u>Count</u>	<u>Support</u>
Dalchini, Py	2	40%
Dalchini, DBMS	2	40%
Py, DBMS	2	40%

Step 3: Candidate Itemset

It occurs only in T3

$$\text{Support} = 1/5 = 20\%$$

Therefore it is not frequent
Effect of max size = 3

The constraint prevents 'Apriori' from generating itemsets containing 4 or more items. Therefore, unnecessary candidate generation and support counting are avoided reducing time, memory usage and computational complexity.

4. University course Hierarchy

Constraint: Generate Rules only at the

Student

Course taken

S1

Engineering, CSE, ML

S2

Engineering, IT

S3

Engineering, CSE

S4

Engineering CSE, ML

S5

Engineering, IT

$$\text{Support} = 2/5 = 40$$

Association Rule

Computer Science

Machine Learning

$$\frac{2}{5} \times 100 = 40\%$$

Thus, The Rule is

Computer Science $\rightarrow$ ML

$$\text{Confidence} = \frac{\text{Support}(S5, ML)}{\text{Support}(CS)}$$

$$\frac{2/5}{2/5} = \frac{2}{2} = 100\%$$

Explanation of course hierarchy

The course hierarchy allows course to be found at different levels such as engineering computer science ML by restricting mining to the computer science level unrelated higher or sibling categories such as IT are excluded. This reduces the search space and produces more specific and meaningful association rules.

5.

Constraint: Age Group = Young

Dataset

Customer	Age Group	Income	Policy
C1	Young	High	Health Ins
C2	Middle	Medium	Vehicle Ins
C3	Young	Low	Travel Ins
C4	Senior	High	Health Ins
C5	Young	Medium	Vehicle Ins

C1 C3 C5 $\rightarrow$ young customers

Meaningful Patterns

1. young + High Income $\rightarrow$ Health Insurance

Support: $\frac{1}{3} = 33.33\%$

2. young + Low Income $\rightarrow$ Travel Insurance

Support: $\frac{1}{3} = 33.33\%$

3. young + Medium Income $\rightarrow$ Vehicle Insurance

Support: $\frac{1}{3} = 33.33\%$

Interpretation

1. High Income young customers $\rightarrow$ Health Insurance
2. Medium - income young customers $\rightarrow$ Vehicle Insurance
3. Low income young customers $\rightarrow$ Travel Insurance